Tata Kelola dan Manajemen Teknologi Informasi

Pengantar Tata Kelola dan Manajemen TI

Team Teaching ISM

S1 Sistem Informasi – Fakultas Rekayasa Industri

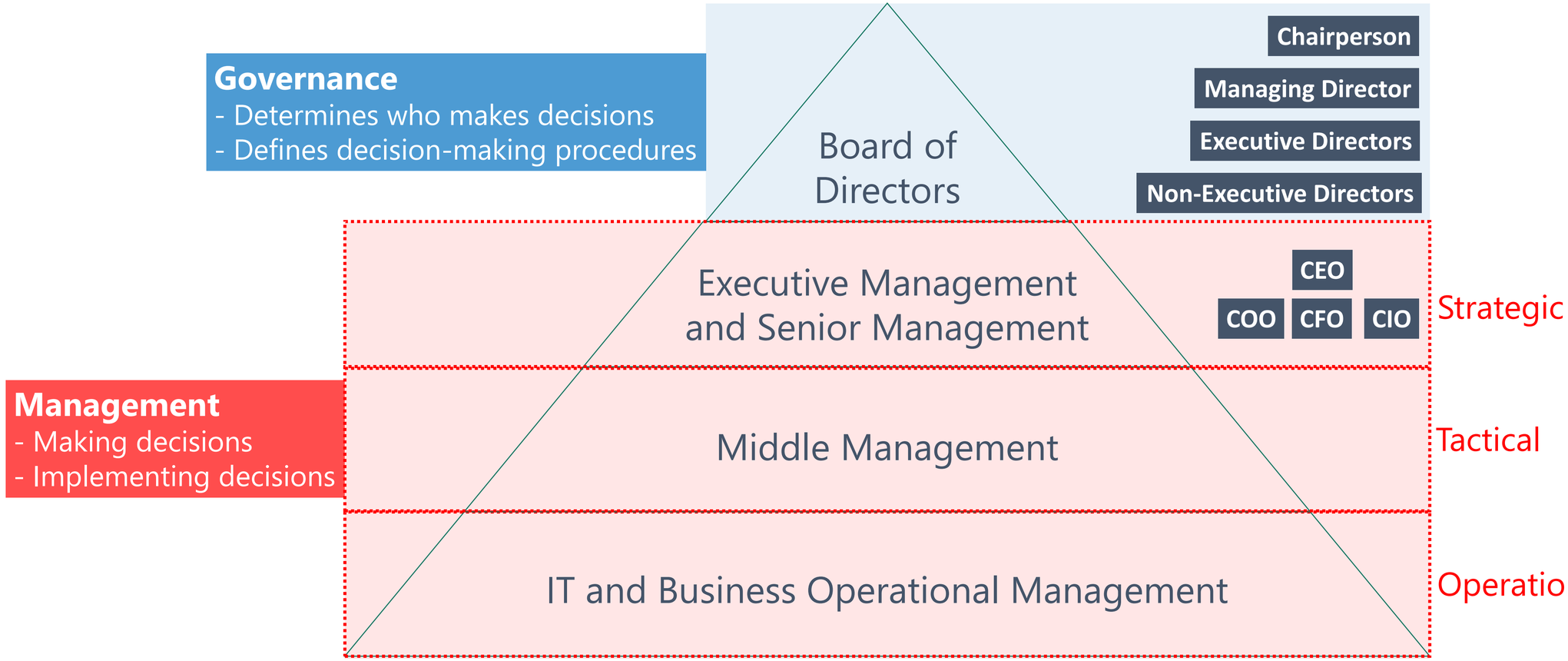
Outline

1. **Definition** of IT Governance and Management
2. **Urgency** of IT Governance and Management
3. IT Governance as **Part of Enterprise Governance**
4. **COBIT** as an IT Governance and Management Framework
5. IT Governance and Management in **COBIT and ISO 38500**
6. COBIT 2019 **Overview**

1. Definition

Who is Responsible for IT?

Let’s take it from the top...



What is IT Governance?

"the responsibility of **executives** and the **board of directors** and consists the **leadership, organisational structures, and processes** that ensure that the enterprise's IT sustains and extends the organisation's strategy and objectives"

Responsibility:

- Executives & Board of Directors

Elements:

- Leadership
- Organizational Structures
- Processes

Objectives:

- Ensure IT sustains and **extends the organization's mission and strategy**

Source: ITGI, 2003

2. Urgency

IT extends the mission and strategy

IT must …

- Deliver **value**
- **Enable** the business

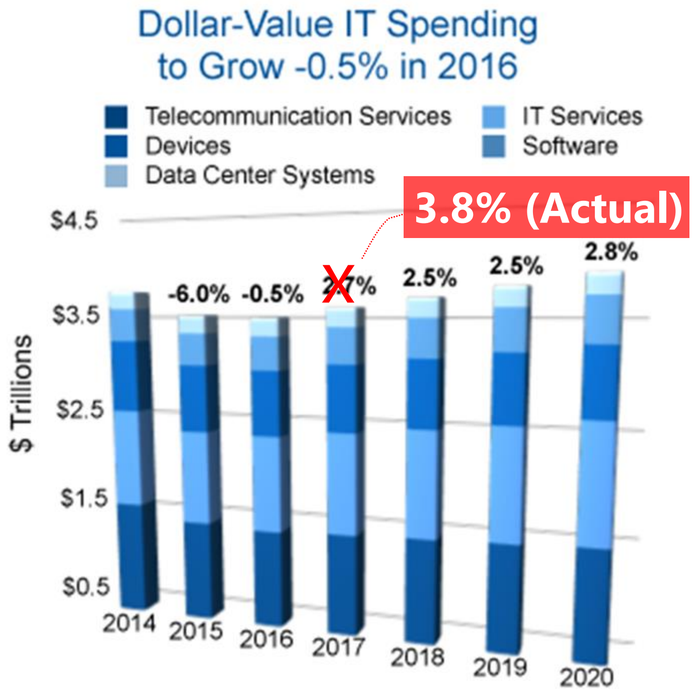


Image Source: Gartner Market Data Book, 1Q16 Update

IT extends the mission and strategy

Fenomena IT paradox

Ketika perusahaan sedang *growth* maupun *flat*, IT membantu *enable growth* maupun *improve margin*, tetapi ketika sedang *declining* dan *budget* IT ikut menurun drastis tetap saja IT sangat dibutuhkan untuk *reduce cost* (efisiensi)

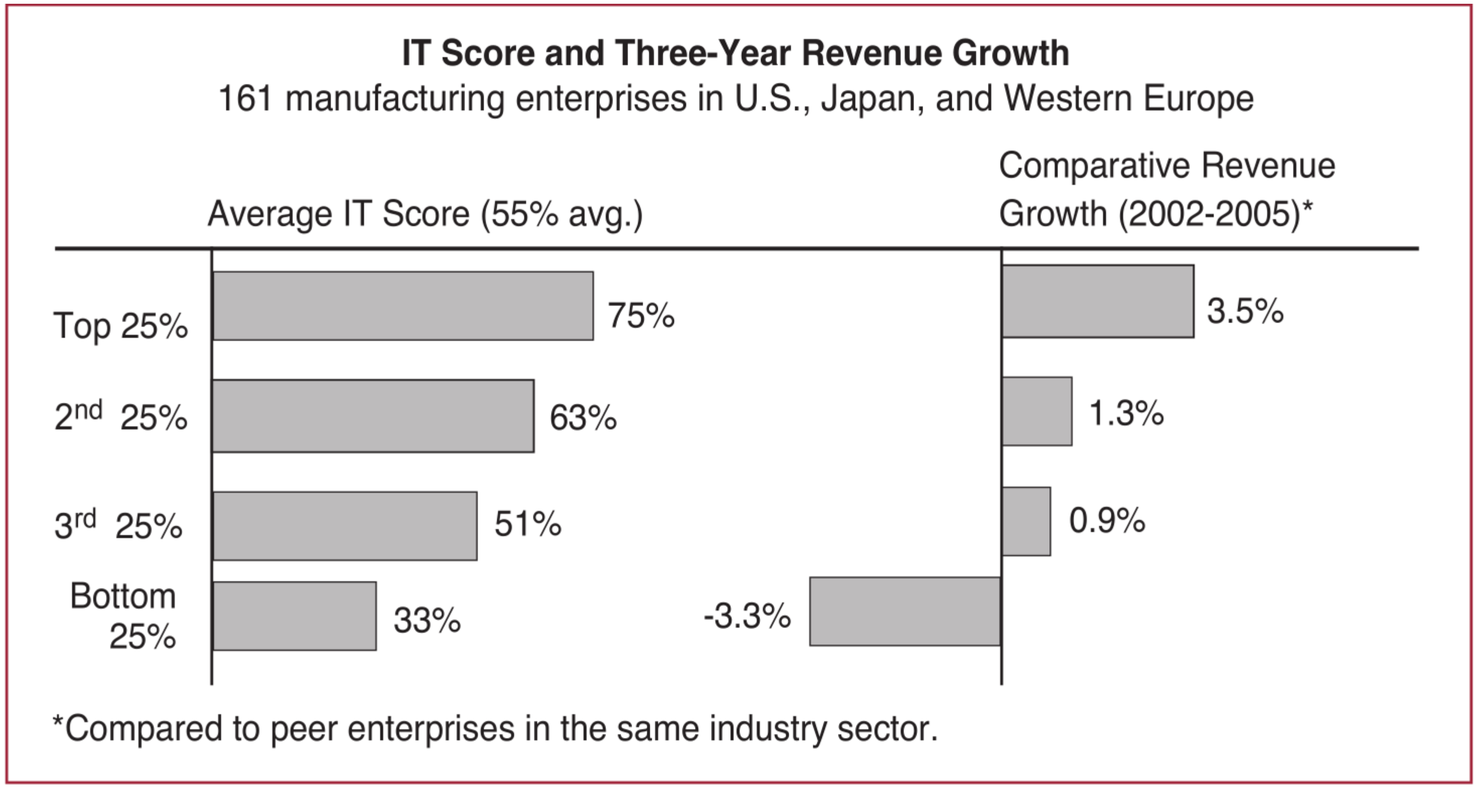
- **IT Productivity Paradox**
  "You can see the computer age everywhere but not in the productivity statistics." (Solow, 1987)
- **IT Investment Paradox / IT Black Hole**
  "large sums are invested in IT and seem to be swallowed by a large black hole without rendering many returns"

IT extends the mission and strategy

Challenges

- **Only 35%** of all **IT project suceeded** while the remainder (65%) were either challenged or failed (The Standish Group Study, 2006)
- **Low return** from **high-cost IT investments**, and **transparency of IT's performance** are two of the top issues
- 40% **do not have good alignment** between **IT plans and business strategy** (ITGI, 2006)

Correlation between IT capabilities and business performance



Enterprises with **better IT capabilities** achieve higher productivity and have greater insights into and control over their businesses, resulting in substantially **higher revenue growth**.

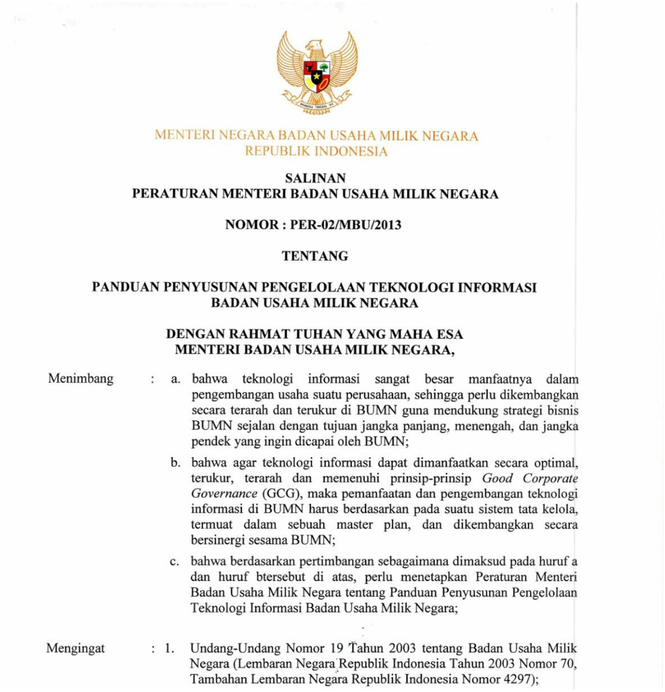
IT matters and that better IT makes a quantifiable, positive difference on business performance.

IT Governance & Management is The Solution

Image Source: Keystone Strategy, Inc., 2006

Regulations as a Driver

Keberadaan regulasi dan peraturan eksternal lainnya menjadi pemicu utama implementasi tata kelola dan pengelolaan TI



MENTERI NEGARA BADAN USAHA MILIK NEGARA
REPUBLIK INDONESIA

SALINAN
PERATURAN MENTERI BADAN USAHA MILIK NEGARA

NOMOR : PER-02/MBU/2013

TENTANG

PANDUAN PENYUSUNAN PENGELOLAAN TEKNOLOGI INFORMASI BADAN USAHA MILIK NEGARA

DENGAN RAHMAT TUHAN YANG MAHA ESA
MENTERI BADAN USAHA MILIK NEGARA,

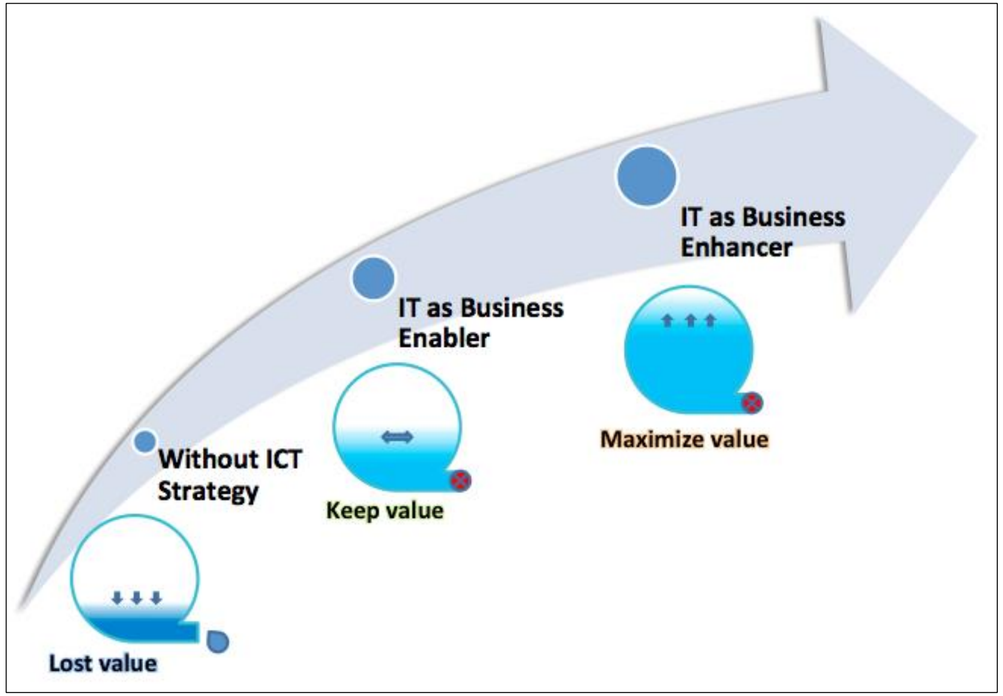
Menimbang :

a. bahwa teknologi informasi sangat besar manfaatnya dalam pengembangan usaha suatu perusahaan, sehingga perlu dikembangkan secara terarah dan terukur di BUMN guna mendukung strategi bisnis BUMN sejalan dengan tujuan jangka panjang, menengah, dan jangka pendek yang ingin dicapai oleh BUMN;

b. bahwa agar teknologi informasi dapat dimanfaatkan secara optimal, terukur, terarah dan memenuhi prinsip-prinsip *Good Corporate Governance* (GCG), maka pemanfaatan dan pengembangan teknologi informasi di BUMN harus berdasarkan pada suatu sistem tata kelola, termuat dalam sebuah master plan, dan dikembangkan secara bersinergi sesama BUMN;

c. bahwa berdasarkan pertimbangan sebagaimana dimaksud pada huruf a dan huruf btersebut di atas, perlu menetapkan Peraturan Menteri Badan Usaha Milik Negara tentang Panduan Penyusunan Pengelolaan Teknologi Informasi Badan Usaha Milik Negara;

Mengingat : 1. Undang-Undang Nomor 19 Tahun 2003 tentang Badan Usaha Milik Negara (Lembaran Negara Republik Indonesia Tahun 2003 Nomor 70, Tambahan Lembaran Negara Republik Indonesia Nomor 4297);



Selain kepatuhan regulasi, tujuan utama Tata Kelola TI adalah memastikan tercapainya peran optimal TI dalam mendukung bisnis dengan mengoptimalkan sumber daya dan risiko untuk mencapai realisasi manfaat TI

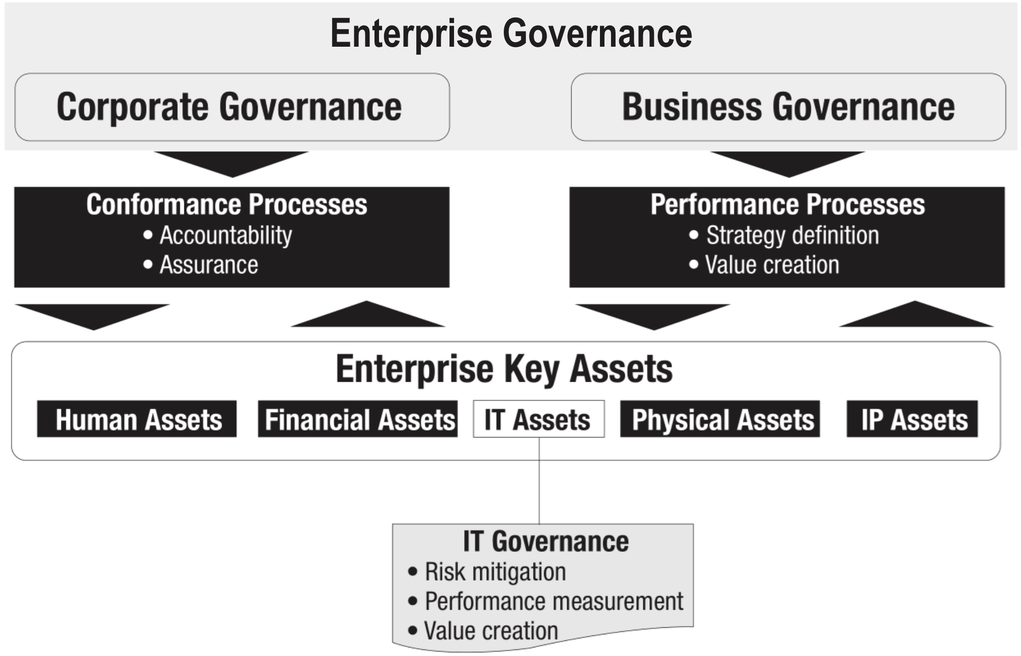
Some of Related Regulations



1. UU Nomor 11 Tahun 2008 tentang **Informasi dan Transaksi Elektronik**
2. PP Nomor 82 Tahun 2012 tentang **Penyelenggaraan Sistem dan Transaksi Elektronik**
3. Permen BUMN No. PER-01/MBU/2011 tentang **Penerapan Tata Kelola Perusahaan yang Baik (GCG) pada BUMN**
4. Permen Kominfo 41/PER/MEN.KOMINFO/11/2007 tentang **Panduan Umum Tata Kelola TIK Nasional**
5. PBI No. 9/15/PBI/2007 tentang **Manajemen Risiko TI untuk Bank Umum**
6. SE No. 9/30/DPNP tentang **Penerapan Manajemen Risiko dalam Penggunaan Teknologi Informasi oleh Bank Umum**
7. Panduan **Penerapan Tata Kelola Keamanan Informasi bagi Penyelenggara Pelayanan Publik** dari Kominfo 2011
8. Permen BUMN PER-02/MBU/2013 tentang **Panduan Penyusunan Pengelolaan Tata Kelola TI BUMN**
9. PTK 53/008 SKKMIGAS/2013 tentang **Pengelolaan TI KKKS**
10. Permen Kominfo No. 4 tahun 2016 tentang **Sistem Manajemen Pengamanan Informasi**
11. ...

3. IT Governance as Part of Enterprise Governance

Part of Enterprise Governance



IT Focus for "Corporate Governance":

- **Information Integrity** (allowing transaction tracking)
- **Increased Timeliness** (speed up decision making)
- **Provision of Reports** (provide for management/leader needs)
- **Automation of Data Collections** (creating financial metrics at corporate level)

IT Focus for "Business Governance":

- **Strategic Alignment** (IT is aligned with the business)
- **Value Delivery** (IT enables the business and maximizes benefits)
- **Resource Management** (IT resources are used responsibly)
- **Risk Management** (IT risks are managed appropriately)
- **Performance Management** (IT processes are measured and monitored)

Image Source: Sandrino-Arndt, 2008

4. COBIT as an IT Governance and Management Framework

Global Survey on IT Governance & Management

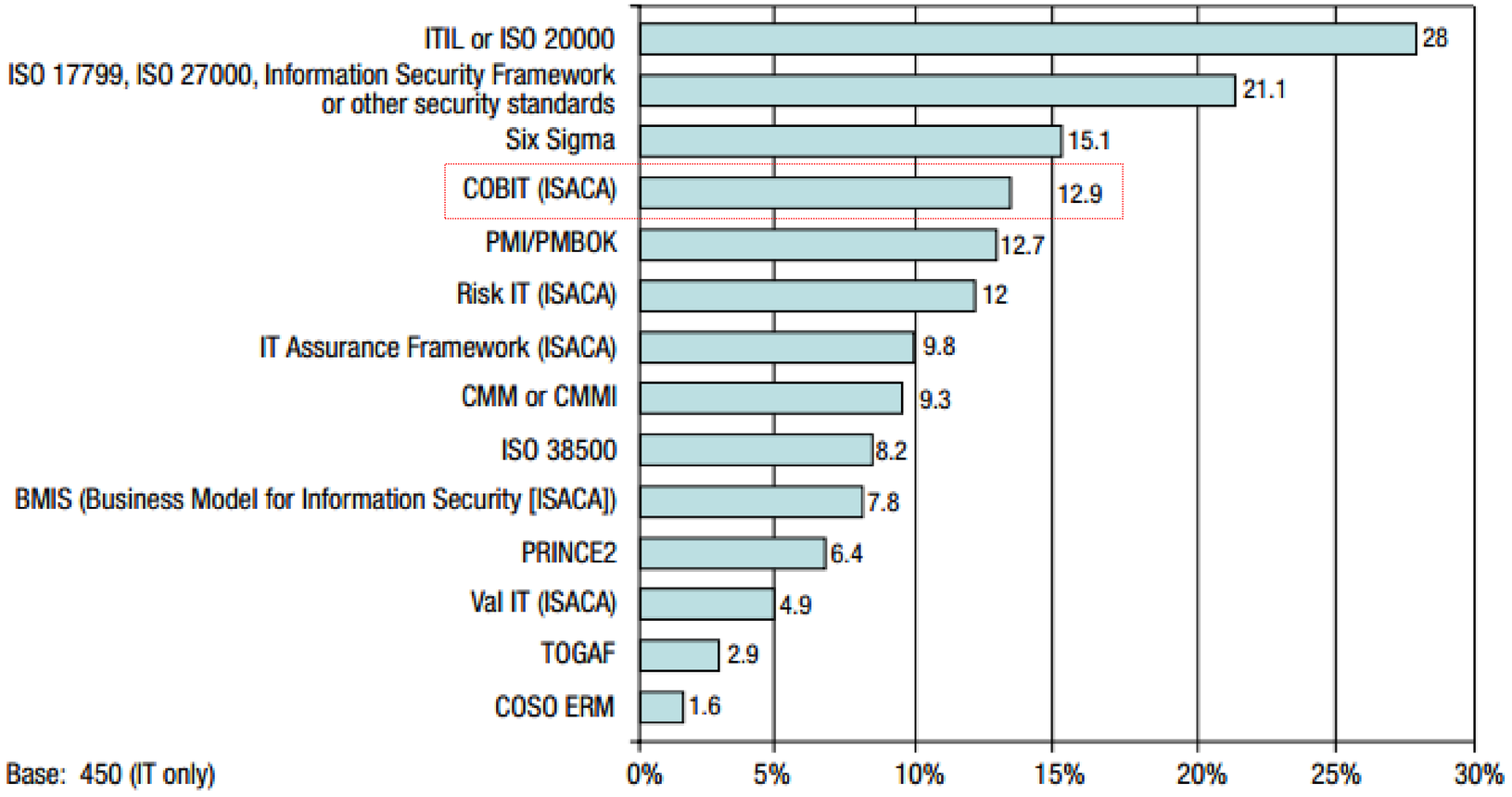


Image Source: ISACA, 2011

COBIT Coverage of Other Frameworks

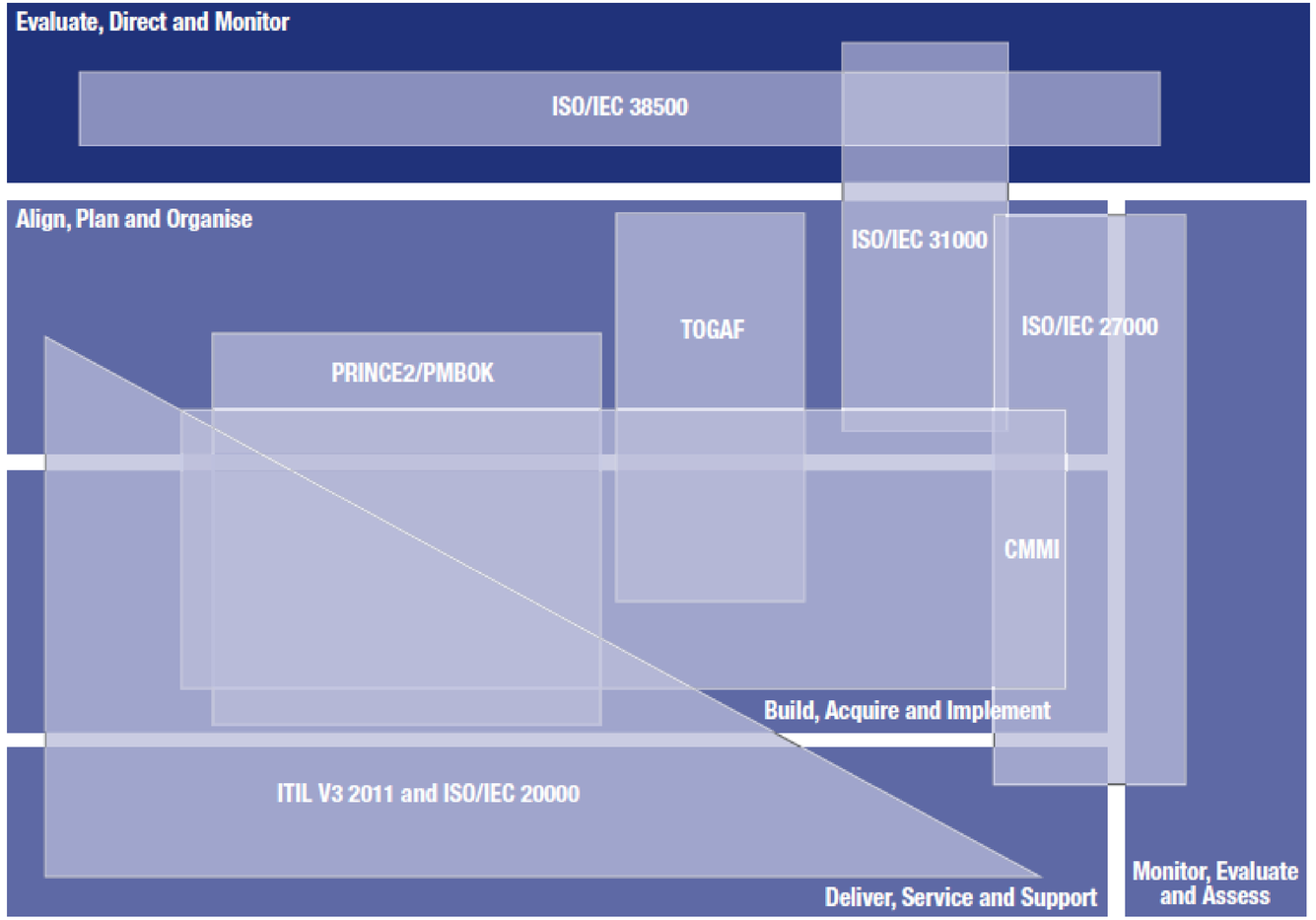


Image Source: ISACA, 2012

Where Does COBIT Fit?

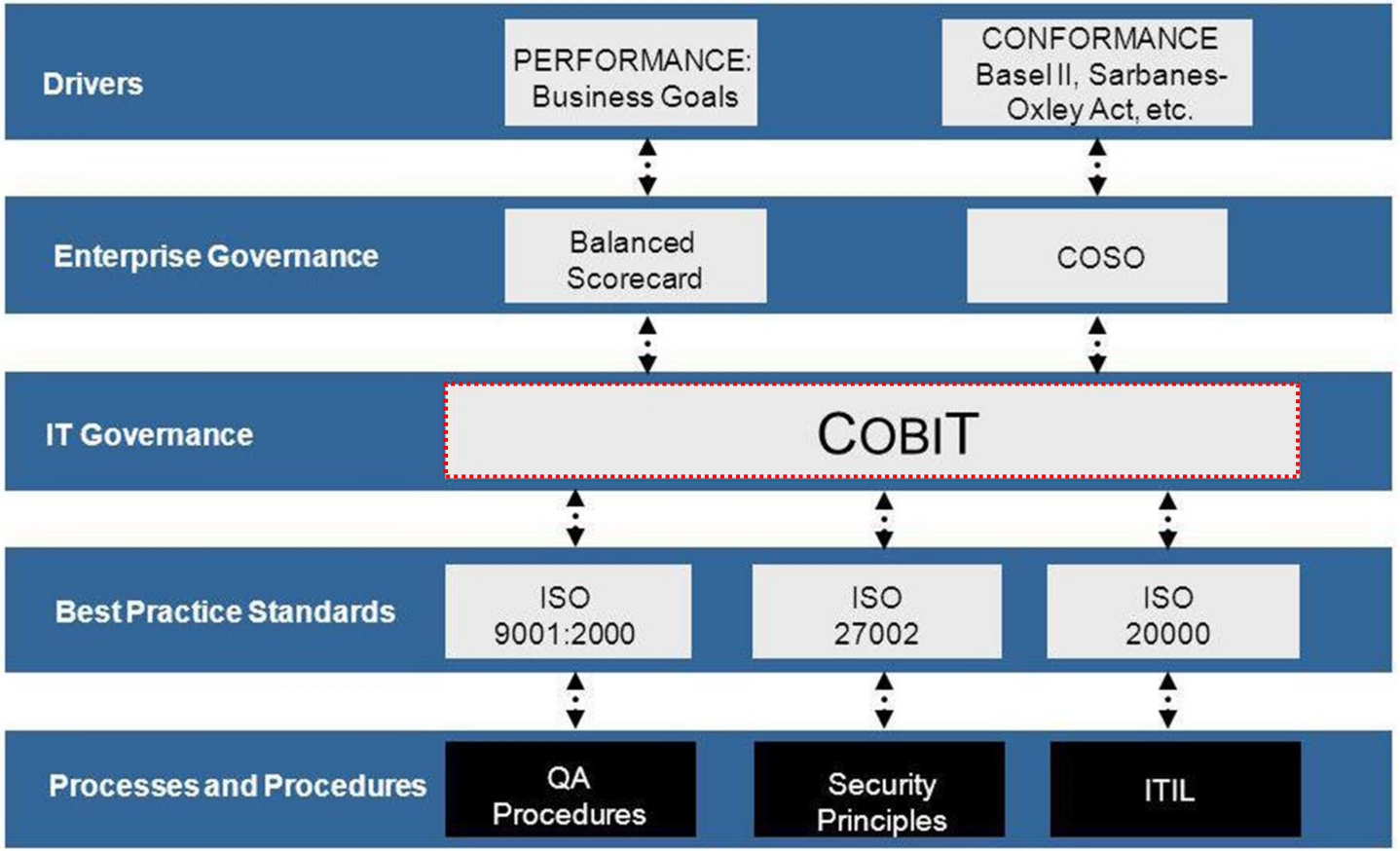
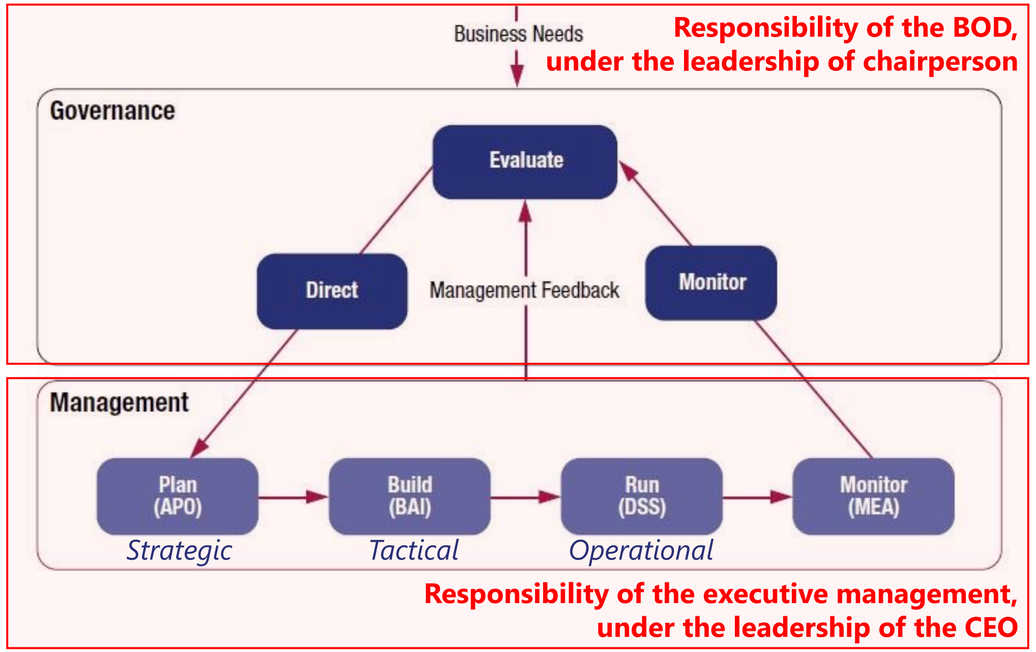


Image Source: ITGI

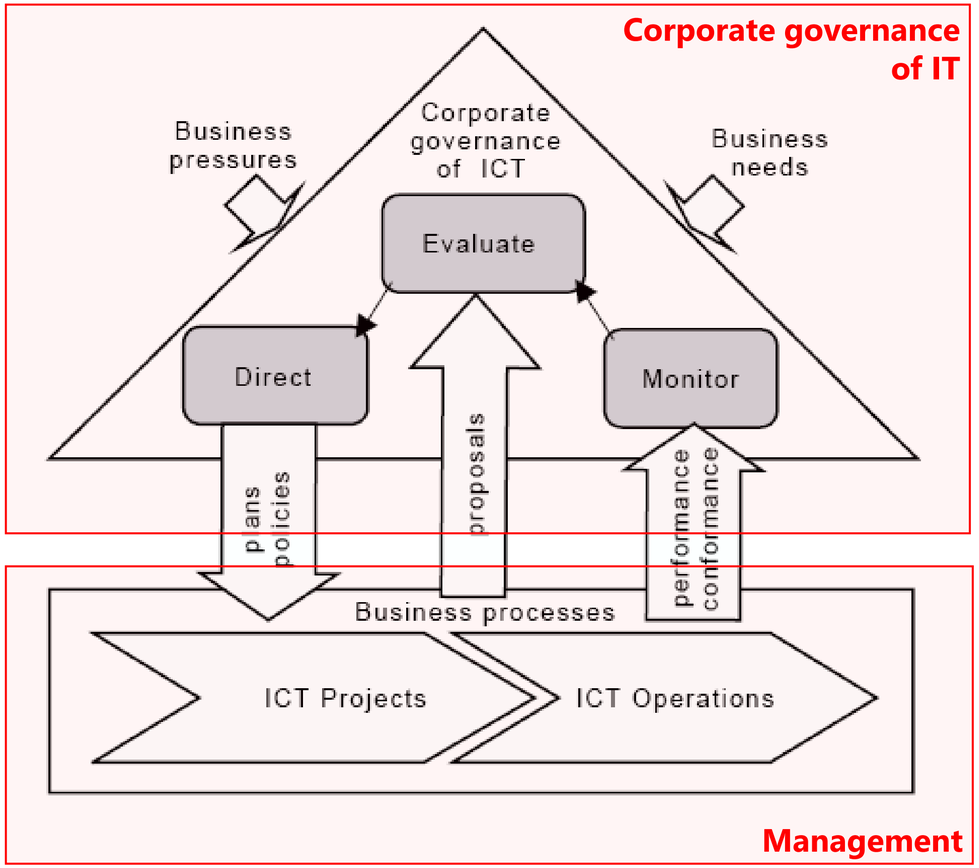
5. COBIT and ISO 38500

COBIT and ISO 38500

COBIT



ISO 38500



ISO 38500

Corporate governance of IT

The system by **which** the **current and future use of IT** is **directed** and **controlled**.

Corporate governance of IT involves evaluating and directing the use of IT to support the organization and monitoring this use to achieve plans. It includes the strategy and policies for using IT within an organization.

Management

The system of **controls and processes required** to **achieve the strategic objectives** set by the organization's governing body. Management is subject to the policy guidance and monitoring set through corporate governance.

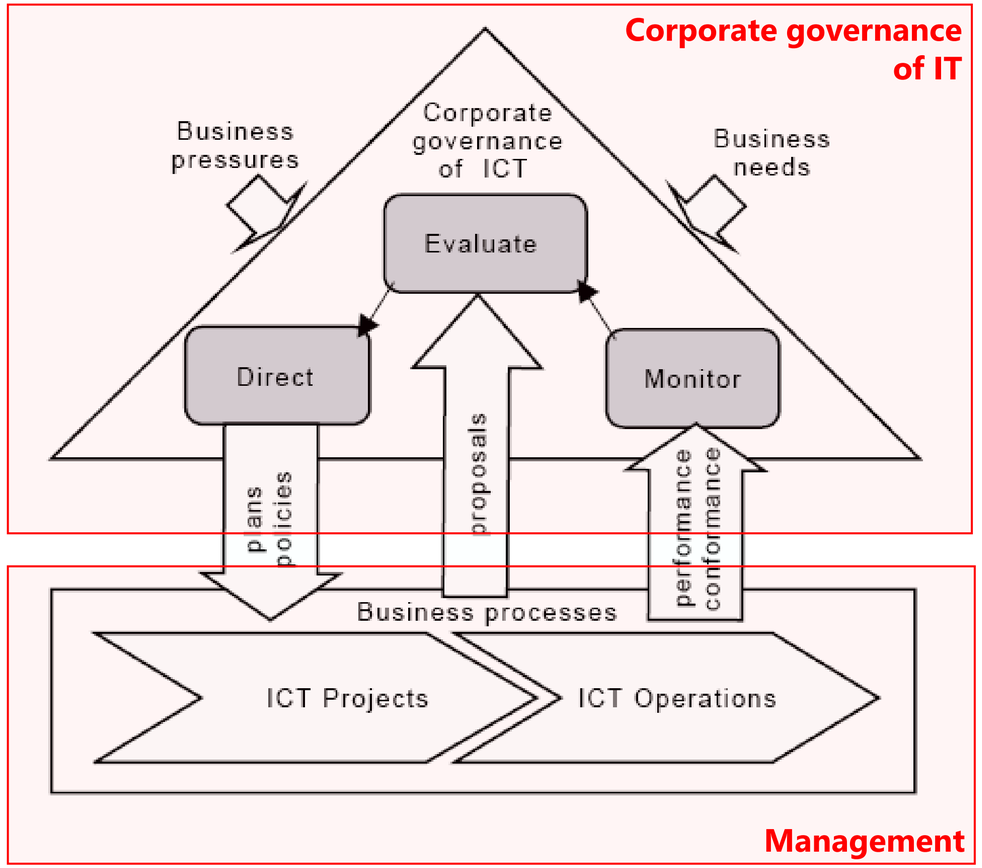
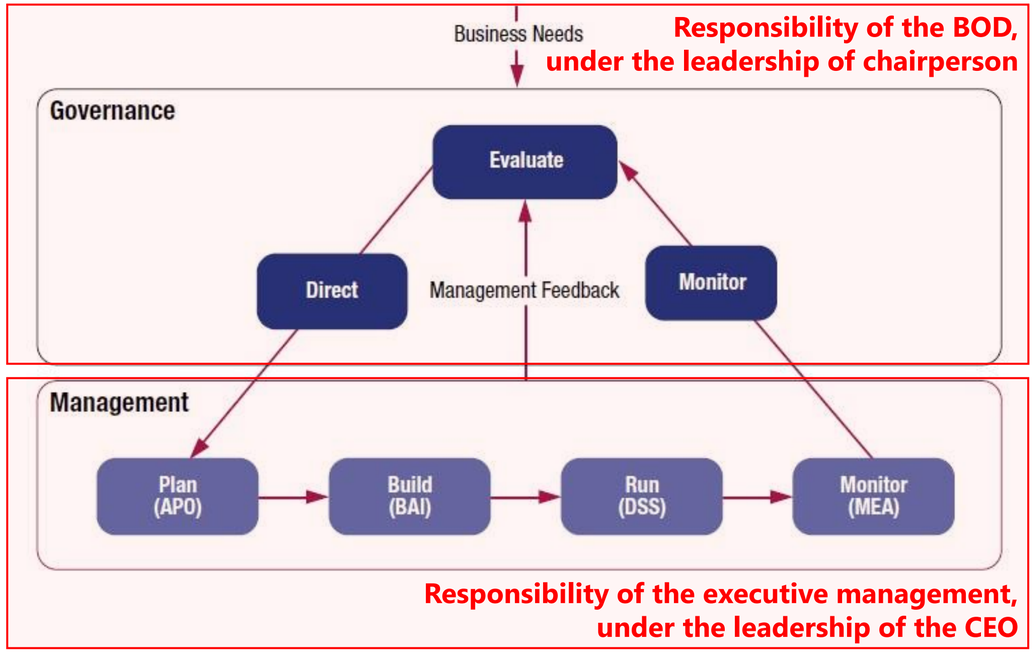


Image Source: ISO, 2008

COBIT

COBIT



COBIT implicitly adopt ISO 38500

Governance ensures that:

- Stakeholder needs, conditions and options are **evaluated** to determine balanced, agreed-on enterprise objectives.
- **Direction** is set through prioritization and decision making.
- Performance and compliance are **monitored** against agreed-on direction and objectives.

Management plans, builds, runs and monitors activities, in alignment with the direction set by the governance body, to achieve the enterprise objectives.

Image Source: ISACA, 2012

6. COBIT 2019 Overview

COBIT Core Model

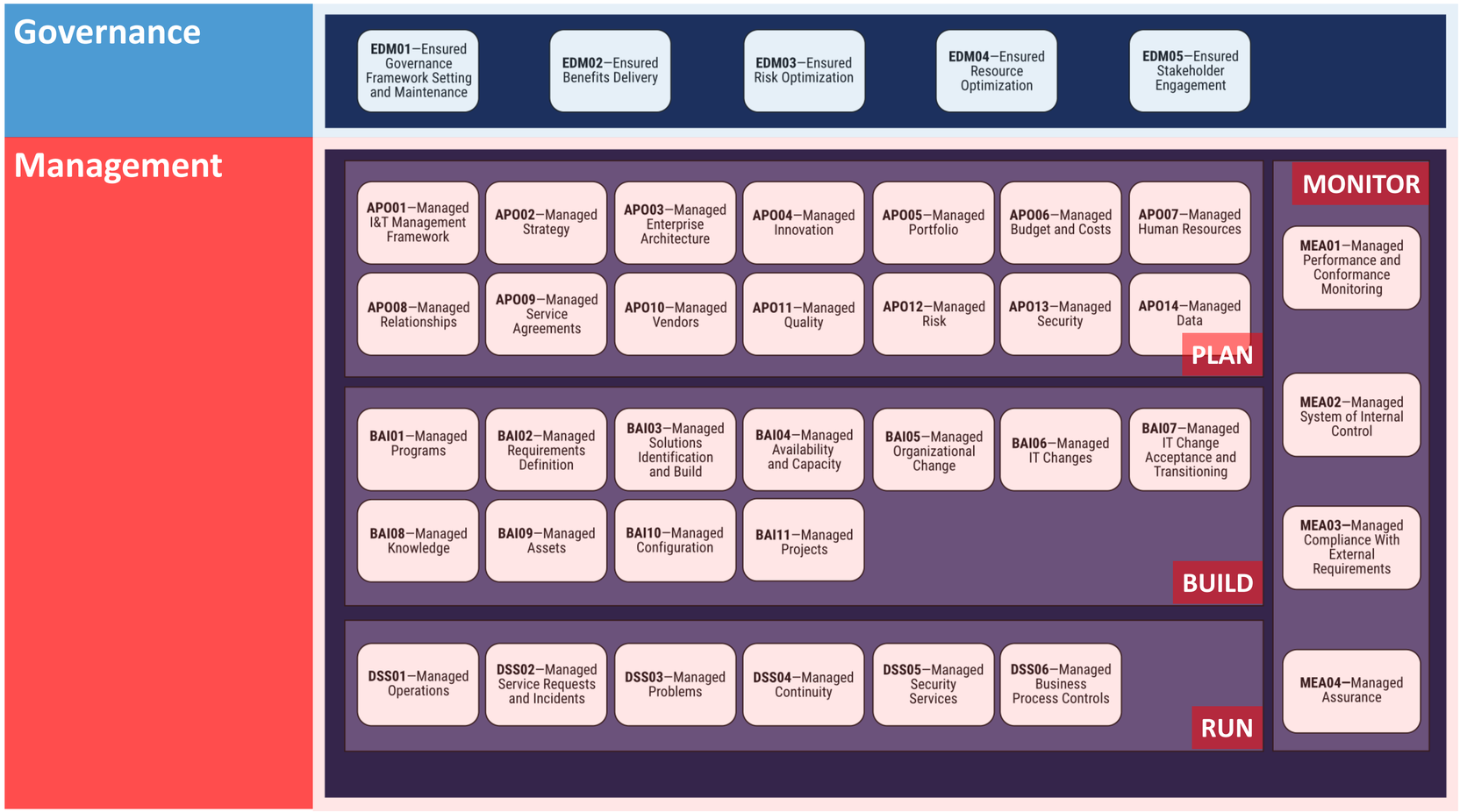
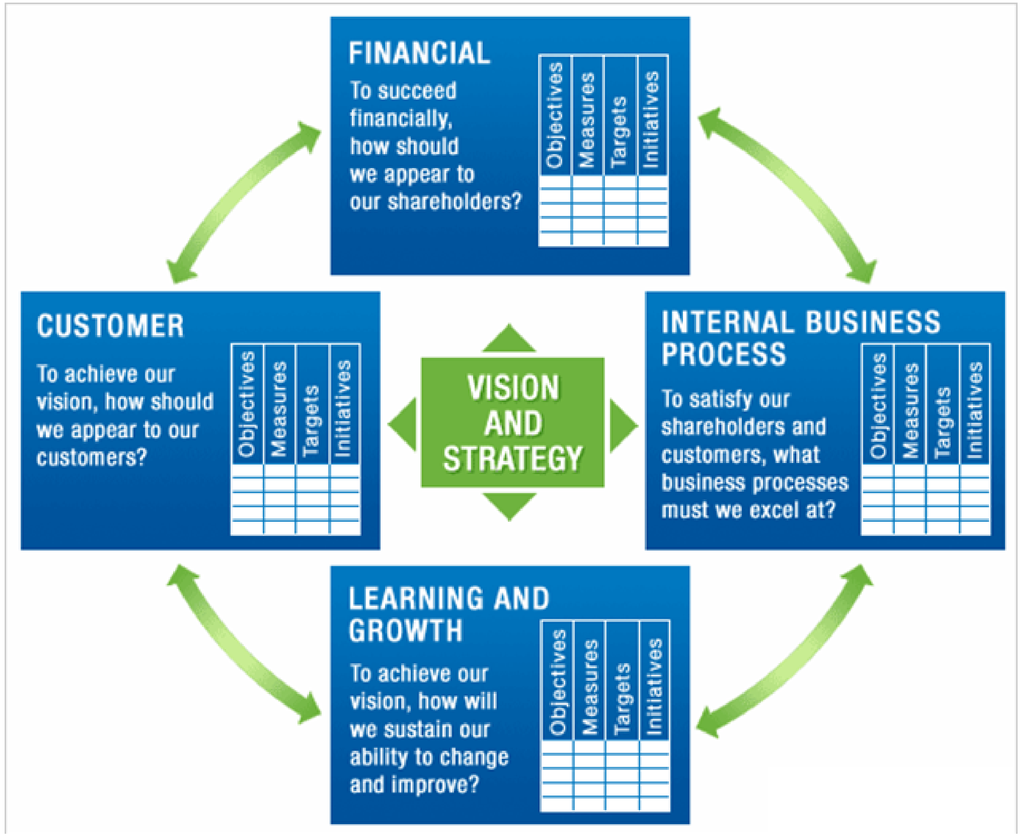


Image Source: ISACA, 2018

Balanced Scorecard Adoption



Cause-and-effect relationship

IF

IT employee's expertise is improved

(learning and growth/future orientation)

THEN

this may result in a better quality of developed systems

(internal business process/operational excellence)

THEN

this may meet better user expectations

(customer/user orientation)

THEN

this may enhance the support of business processes

(financial/business contribution)

COBIT Goals Cascade

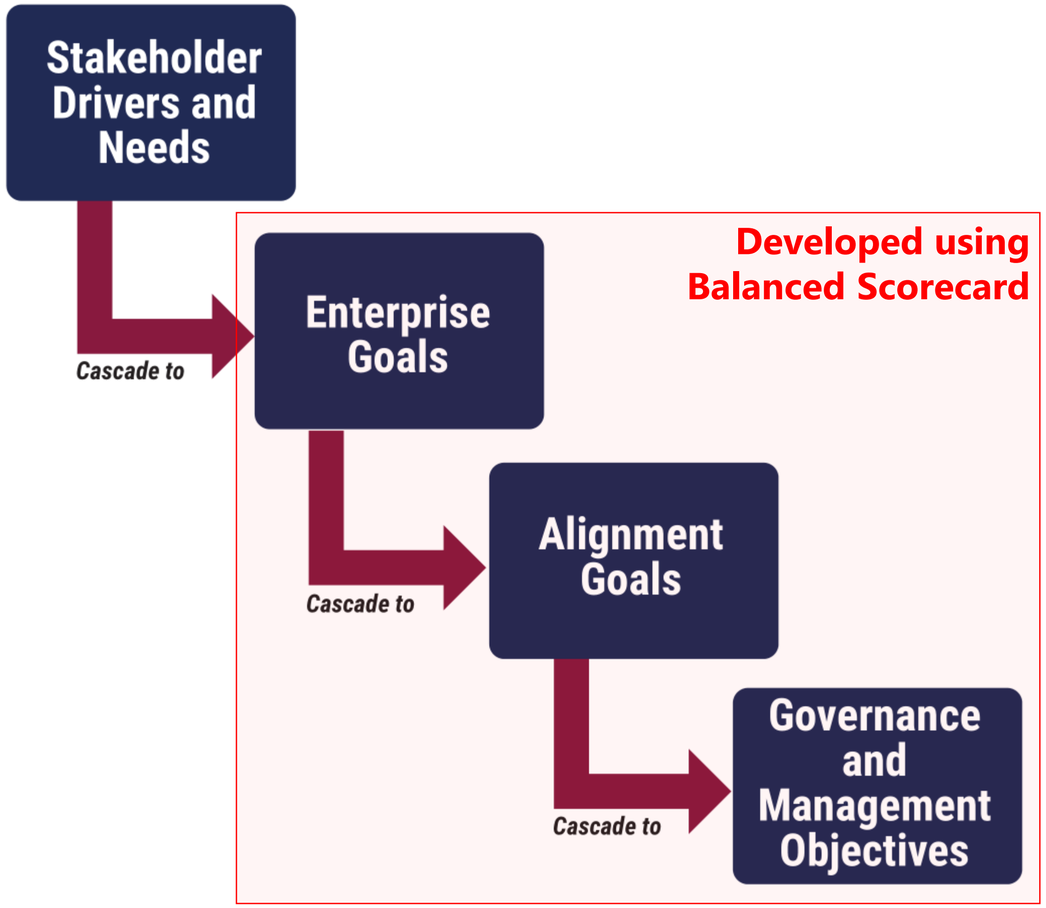


Image Source: ISACA, 2018

COBIT 2019 Enterprise Goals

BSC dimension	Ref.	Enterprise Goal
Financial	EG01	Portfolio of competitive products and services
	EG02	Managed business risk
	EG03	Compliance with external laws and regulations
	EG04	Quality of financial information
Customer	EG05	Customer-oriented service culture
	EG06	Business-service continuity and availability
	EG07	Quality of management information
Internal	EG08	Optimization of internal business process functionality
	EG09	Optimization of business process costs
	EG10	Staff skills, motivation and productivity
	EG11	Compliance with internal policies
Learning and Growth	EG12	Managed digital transformation programs
	EG13	Product and business innovation

COBIT 2019 Alignment Goals

BSC dimension	Ref.	Alignment Goal
Financial	AG01	IT compliance and support for business compliance with external laws and regulations
	AG02	Managed IT-related risk
	AG03	Realized benefits from IT enabled investments and services portfolio
	AG04	Quality of technology-related financial information
Customer	AG05	Delivery of I&T services in line with business requirements
	AG06	Agility to turn business requirements into operational solutions
Internal	AG07	Security of information, processing infrastructure and applications, and privacy
	AG08	Enabling and supporting business processes by integrating applications and technology
	AG09	Delivery of programs on time, on budget and meeting requirements and quality standards
	AG10	Quality of IT management information
	AG11	IT compliance with internal policies
Learning and Growth	AG12	Competent and motivated staff with mutual understanding of technology and business
	AG13	Knowledge, expertise and initiatives for business innovation

Mapping Enterprise and Alignment Goals

		EG01	EG02	EG03	EG04	EG05	EG06	EG07	EG08	EG09	EG10	EG11	EG12	EG13
		Portfolio of competitive products and services	Managed business risk	Compliance with external laws and regulations	Quality of financial information	Customer-oriented service culture	Business service continuity and availability	Quality of management information	Optimization of internal business process functionality	Optimization of business process costs	Staff skills, motivation and productivity	Compliance with internal policies	Managed digital transformation programs	Product and business innovation
AG01	I&T compliance and support for business compliance with external laws and regulations		S	P								S		
AG02	Managed I&T-related risk		P				S							
AG03	Realized benefits from I&T-enabled investments and services portfolio	S				S			S	S			P	
AG04	Quality of technology-related financial information				P			P		P				
AG05	Delivery of I&T services in line with business requirements	P				S	S		S				S	
AG06	Agility to turn business requirements into operational solutions	P				S			S				S	S
AG07	Security of information, processing infrastructure and applications, and privacy		P				P							
AG08	Enabling and supporting business processes by integrating applications and technology	P				P			S		S		P	S
AG09	Delivering programs on time, on budget and meeting requirements and quality standards	P				S			S	S			P	S
AG10	Quality of I&T management information				P			P		S				
AG11	I&T compliance with internal policies		S	P								P		
AG12	Competent and motivated staff with mutual understanding of technology and business					S					P			
AG13	Knowledge, expertise and initiatives for business innovation	P		S									S	P

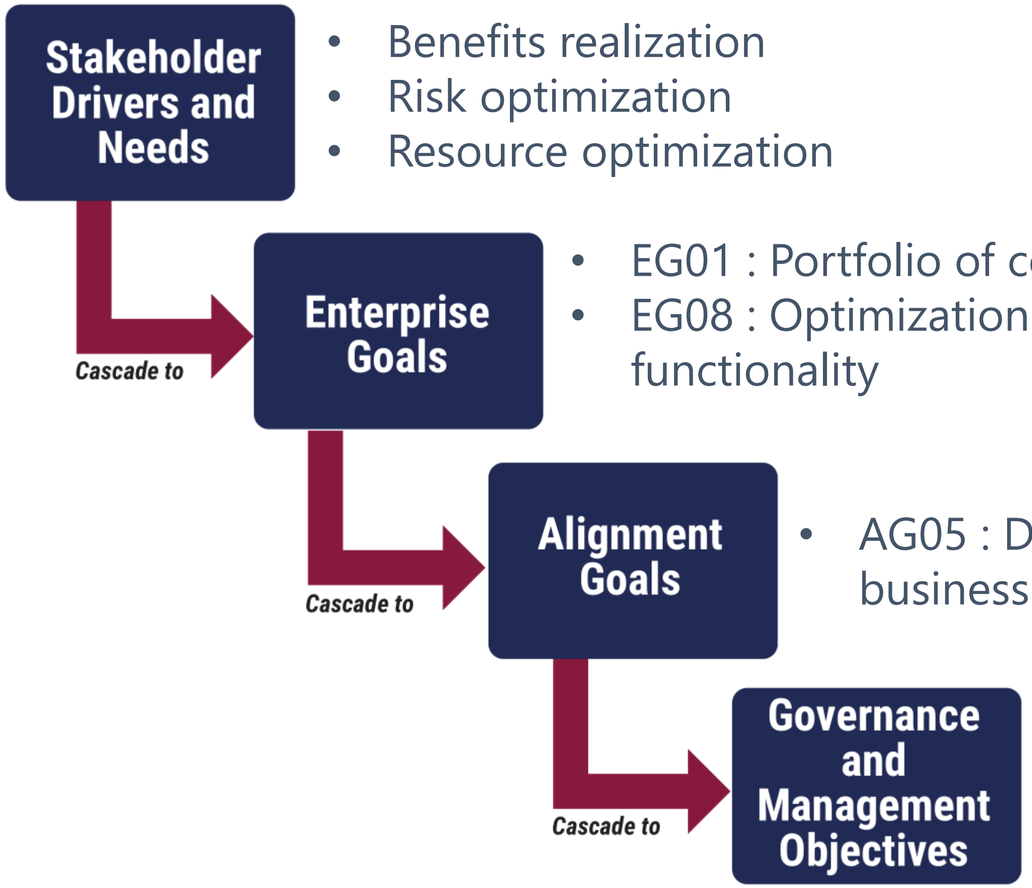
Mapping Alignment Goals and Objectives

		AG01	AG02	AG03	AG04	AG05	AG06	AG07	AG08	AG09	AG10	AG11	AG12	AG13
		I&T compliance and support for business compliance with external laws and regulations	Managed I&T-related risk	Realized benefits from I&T-enabled investments and services portfolio	Quality of technology-related financial information	Delivery of I&T services in line with business requirements	Agility to turn business requirements into operational solutions	Security of information, processing infrastructure and applications, and privacy	Enabling and supporting business processes by integrating applications and technology	Delivering programs on time, on budget and meeting requirements and quality standards	Quality of I&T management information	I&T compliance with internal policies	Competent and motivated staff with mutual understanding of technology and business	Knowledge, expertise and initiatives for business innovation
EDM01	Ensured governance framework setting and maintenance	P	S	P					S			S		
EDM02	Ensured benefits delivery			P		S	S		S					S
EDM03	Ensured risk optimization	S	P					P				S		
EDM04	Ensured resource optimization			S		S	S		S	P			S	
EDM05	Ensured stakeholder engagement				S						P	S		
APO01	Managed I&T management framework	S	S	P		S		S	S	S	S	P		
APO02	Managed strategy			S		S	S		P				S	S
APO03	Managed enterprise architecture			S		S	P	S	P					
APO04	Managed innovation			S			P		S				S	P
APO05	Managed portfolio			P		P	S		S	S				
APO06	Managed budget and costs			S	P					P	S			
APO07	Managed human resources			S		S				S			P	P
APO08	Managed relationships			S		P	P		S	S			P	P
APO09	Managed service agreements					P			S					
APO10	Managed vendors					P	S			S				
APO11	Managed quality			S	S	S				P	P			
APO12	Managed risk		P					P						
APO13	Managed security	S	S					P						
APO14	Managed data	S	S		S			S			P			

Mapping Alignment Goals and Objectives

		AG01	AG02	AG03	AG04	AG05	AG06	AG07	AG08	AG09	AG10	AG11	AG12	AG13
		I&T compliance and support for business compliance with external laws and regulations	Managed I&T-related risk	Realized benefits from I&T-enabled investments and services portfolio	Quality of technology-related financial information	Delivery of I&T services in line with business requirements	Agility to turn business requirements into operational solutions	Security of information, processing infrastructure and applications, and privacy	Enabling and supporting business processes by integrating applications and technology	Delivering programs on time, on budget and meeting requirements and quality standards	Quality of I&T management information	I&T compliance with internal policies	Competent and motivated staff with mutual understanding of technology and business	Knowledge, expertise and initiatives for business innovation
BAI01	Managed programs			P			S		S	P				
BAI02	Managed requirements definition			S		P	P		S	P			S	
BAI03	Managed solutions identification and build			S		P	P		S	P				
BAI04	Managed availability and capacity					P		S		S				
BAI05	Managed organizational changes			P		S	S		P	P			S	
BAI06	Managed IT changes		S			S	P		S					
BAI07	Managed IT change acceptance and transitioning		S				P			S				
BAI08	Managed knowledge			S			S		S	S			P	P
BAI09	Managed assets				P						S			
BAI10	Managed configuration					S		P						
BAI11	Managed projects			P		S	P			P				
DSS01	Managed operations					P			S					
DSS02	Managed service requests and incidents		S			P		S						
DSS03	Managed problems		S			P		S						
DSS04	Managed continuity		S			P		P						
DSS05	Managed security services	S	P			S		P				S		
DSS06	Managed business process controls		S			S		S	P			S		
MEA01	Managed performance and conformance monitoring	S		S		P				S	P	S		
MEA02	Managed system of internal control	S	S		S	S		S		S	S	P		
MEA03	Managed compliance with external requirements	P										S		
MEA04	Managed assurance	S	S		S	S		S			S	P		

Example of Goals Cascade



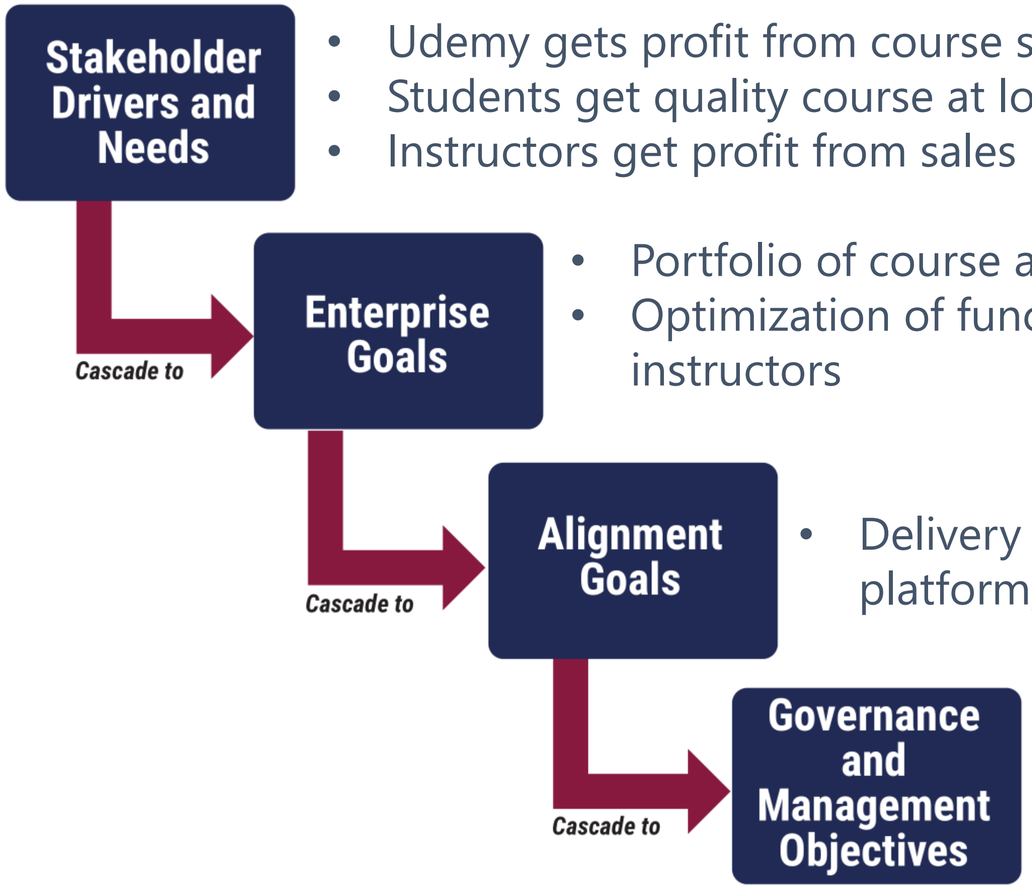
- Benefits realization
- Risk optimization
- Resource optimization

- EG01 : Portfolio of competitive products and services
- EG08 : Optimization of internal business process functionality

- AG05 : Delivery of I&T services in line with business requirements

- DSS02 : Managed service requests and incidents

Example of Goals Cascade: Udemy



- Udemy gets profit from course sales
- Students get quality course at low prices
- Instructors get profit from sales

- Portfolio of course at competitive quality and prices
- Optimization of functionality to Udemy, students and instructors

- Delivery of I&T services through Udemy platform, website and mobile applications

- Customer support to managed a Service requests and incidents

Thank You

That’s all